

Yongjun Rong

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Cherry Hill, NJ, USA, 08003 | US Citizen

Summary

A Principal Platform Engineer with deep expertise in building scalable backend systems, automation frameworks, and infrastructure control planes from the ground up. I specialize in designing and implementing robust platforms using Go, Python, and Terraform, with extensive experience in Kubernetes, CI/CD automation, and developer tooling. My work focuses on creating systems that manage complex relationships across infrastructure, embedding business logic into operational workflows, and modernizing primitive structures into centralized, automated platforms. I bring a builder's mindset with strong programming proficiency and experience designing internal APIs and automation frameworks for enterprise-scale infrastructure.

Tags: Platform Engineering, Backend Systems, Automation

Skills

Scalable Backend Systems: Proven experience designing and building scalable systems from scratch that automate complex infrastructure structures and manage inventory relationships across enterprise environments.

Platform Engineering & Automation: Expertise in creating automation frameworks, infrastructure control planes, and developer tooling using Go, Python, and Terraform. Experience designing internal APIs and modernizing primitive structures into robust platforms.

Infrastructure as Code & Kubernetes: Deep familiarity with Kubernetes, Terraform, Terra-grunt, and infrastructure-as-code tools. Experience building and managing containerized platforms on EKS/AKS/GKE with CI/CD automation.

Programming & API Design: Strong programming proficiency in Go and Python, with experience in Java/Kotlin. Proven ability to design and implement internal APIs, microservices, and automation frameworks for infrastructure teams.

Enterprise Platform Architecture: Extensive experience designing and implementing platform solutions at enterprise scale, with understanding of SRE principles, metrics-driven observability, and reliability engineering practices.

Technologies

Programming Languages: Go, Python, Java, Terraform, Bash

Platform Tools: Kubernetes/EKS, Terraform/Terragrunt, AWS Lambda/SQS/SNS/EventBridge, CI/CD (GitLab, Jenkins, Bamboo), Docker, Prometheus/Grafana

Work Experience

Director of Cloud Architecture

Feb. 2025 – Current

Biophy, Cherry Hill, NJ (WFH), USA

Architected and implemented comprehensive infrastructure as code (IaC) from the ground up, designing scalable automation frameworks and a custom AWS control plane to manage complex cloud resources. Led the development of a data ingestion system, applying platform engineering principles to create

robust, automated workflows.

Technologies: Go/Python, Terraform/Terragrunt, AWS (Lambda, SQS, SNS, EventBridge), Infrastructure as Code, API Design, Kubernetes, Automation Frameworks

Project: Comprehensive Infrastructure as Code (IaC) Design and Implementation

- Designed and implemented modular Terraform configurations with reusable modules for consistent deployment across environments.
- Architected a scalable automation framework for importing, cataloging, and managing AWS resources (EC2, VPC, IAM, Security Groups).
- Established comprehensive security controls and automated validation procedures for infrastructure changes.
- Created a custom alerting system leveraging Lambda, SQS, and SNS as part of the infrastructure control plane.

Senior Cloud Engineer II (Associate)

Dec. 2021 – Feb. 2025

McKinsey & Company, Cherry Hill, NJ (WFH), USA

Built scalable genAI systems and automation platforms to accelerate mainframe migration. Developed foundational services that managed complex data relationships and transformation pipelines at enterprise scale.

Technologies: Python/Go, Kubernetes/EKS, Terraform, Microservices, API Design, EventBridge/Lambda/Step Functions, CI/CD Automation

Project: Migration/Refactoring of Mainframe EDA System to AWS

- Developed SpringBoot EKS router to convert XML/XSD to JSON schema and integrate with MQ events.
- Designed and implemented reusable monitoring Terraform module for all AWS resources using CloudWatch metrics.
- Built reusable report generator using Lambda and EventBridge deployed as microservices.
- Created automation framework for running custom AWS CloudWatch insights queries on custom schedules.

Staff Engineer

Dec. 2020 – Dec. 2021

Yotascale, Cherry Hill, NJ (WFH), USA

Led the architecture and development of a comprehensive cloud cost analytics platform, building scalable data pipelines, analytics microservices, and automation frameworks for AWS, Azure, and GCP.

Technologies: Java/Kotlin/Python, Kubernetes/EKS, Terraform/Terragrunt, Microservices, API Design, Automation Frameworks, CI/CD

Project: Lead Architect for Cloud Cost Analytics/Optimization/Management Platform

- Introduced and built Microservice Design First Methodology based on OpenAPI/Swagger Specification.
- Designed and developed Cost Analytics/Lens Management Microservices using SpringBoot with ClickHouse and PostgreSQL.
- Implemented bitbucket CI/CD pipeline to build, deploy Microservices/GraphQL to EKS with Istio.
- Designed/Developed multiple Terraform modules with Terragrunt to maintain dependencies between modules.

- Developed Terraform templates to dynamically create S3 buckets with IAM policies for k8s service account access control using OIDC.

Technical Infrastructure/Devops Architect

Mar. 2020 – Nov. 2021

Gigster Inc., Cherry Hill, NJ, USA

Designed and implemented comprehensive AWS infrastructure architecture including CI/CD automation, Terraform infrastructure as code, and EKS container orchestration for microservices and distributed systems.

Technologies: Terraform/Terragrunt, Kubernetes/EKS, Python/Go, Automation, CI/CD, Infrastructure as Code

Project: Images Manipulation/Detection Software Evaluation System

- Designed and implemented fully automated AWS stack using terraform/terragrunt/bash.
- Developed Terraform scripts/modules to dynamically deploy multiple microservices to EKS cluster.
- Developed Terraform scripts to deploy/configure AWS ALB as EKS ingress controller using helm provider.
- Designed/Developed multiple Terraform modules with Terragrunt to maintain dependencies between modules.

Education

Master of Computer Science

Sept. 2002 – Jun. 2006

Texas Tech University, Lubbock, TX, USA

GPA: 4.0/4.0

Languages

English: Fluent

Chinese: Native

Hobbies

Playing with cutting-edge technologies: Docker, Go, Kafka, Kubernetes, Terraform, Infrastructure Automation

IoT or Robot programming: Programming against iRobot create SDK and MIPOSaur SDK, building automation systems.

Personalized JSON Schema Resume:

- Adapted the JSON schema and developed new data filtering utility methods in JavaScript.
- Built automation for PDF generation from HTML using Puppeteer.
- Developed custom CSS3 styles and print formatting automation.